

Standard Perfusate Components, Volume, and Function

- **Red blood cells (hematocrit 60%):** 840 mL. Function: Oxygen carrier. Derived from the blood bank and dissolved in SAGM (saline, adenine, glucose, mannitol) solution
- **Gelofusine 4%:** 500 mL. Function: Provides physiological colloid osmotic pressure and osmolality
- **Albumin 20%:** 250 mL. Function: Provides physiological colloid osmotic pressure
- **Total parenteral nutrition (TPN N14G30E):** 20 mL. Function: Metabolic substrate provision
- **Sterile water:** 335 mL. Function: Solvent volume expansion
- **Addamel:** 10 mL. Function: Trace elements supplementation
- **Metronidazol (Flagyl) 5 mg/ml:** 44 mL. Function: Antimicrobial prophylaxis
- **Insulin (NovoRapid) 100 IU/ml:** 1 mL. Function: Glycemic regulation
- **Cernevit:** 2 mL. Function: Multi-vitamin supplementation
- **Heparin (5000 IU/ml):** 2 mL. Function: Anticoagulation
- **Cefazolin 1g/5mL:** 2 mL. Function: Antimicrobial prophylaxis
- **Taurocholic acid sodium salt 0.1% (1mg/mL):** 7.7 mL. Function: Exogenous bile acid provision
- **Sodium bicarbonate 8.4%:** 35 mL. Function: pH buffering
- **KCl 1 mmol/mL:** 2 mL. Function: Potassium baseline establishment
- **Glutathion (Tationil) 600mg/4mL:** 14 mL. Function: Antioxidant protection
- *Total Initial Volume:* 2064.7 mL

Calculated NMP Perfusate Formulation (Laboratory Substitute Protocol)

- **Red Blood Cells (AS-1, Leukocyte Reduced): 2 Units (approx 650 mL).** Function: Oxygen carrier
- **6% Hetastarch in 0.9% NaCl: 333 mL (Mass-matched substitution for Gelofusine 4%).** Function: Primary synthetic colloid osmotic pressure provision
- **0.9% Normal Saline (NaCl): 677 mL.** Function: Volume expansion and stoichiometric sodium baseline establishment.
- **Albumin 20%: 250 mL.** Function: Natural colloid osmotic pressure provision
- **CLINIMIX E 5/20: 30 mL.** Function: Metabolic substrate provision
- **Vitamin B Complex HP: 5 mL.** Function: Water-soluble vitamin supplementation
- **Meropenem (500 mg): Reconstituted in 10 mL 0.9% NaCl.** Function: Broad-spectrum antimicrobial prophylaxis
- **Cefuroxime (750 mg): Reconstituted in 10 mL 0.9% NaCl.** Function: Broad-spectrum antimicrobial prophylaxis
- **L-Glutathione (Reduced, 97%): 2000 mg dissolved in 20 mL 0.9% NaCl.** Function: Antioxidant protection
- **Sodium Bicarbonate 8.4%: 35 mL.** Function: pH buffering
- **Heparin (5000 IU/ml): 2 mL.** Function: Anticoagulation
- **KCl (1 mmol/mL): 2 mL. Hyperkalemia (Remove when use expired RBCs)**
- **Insulin (100 IU/ml): 1 mL.** Function: Glycemic regulation
- **Calcium Gluconate 10%: 10 mL at COR initiation.** Function: Essential cofactor provision for bile production
- **Total Estimated Volume: approx 1985 mL.**

Calculated Ionic Concentrations

- Sodium (Na^+): approx 133 mmol/L.
- Chloride (Cl^-): approx 102.4 mmol/L.
- Calcium (Ca^{2+}): approx 1.14 mmol/L (following the initial 10 mL bolus).

Perfusate Correction Guide and Rationale

- **Anticoagulation (Heparin 5000 IE/ml):**
 - *Protocol:* Administer 0.2 mL (1000 units) at the start of NMP. Administer 0.2 mL every hour thereafter
 - *Rationale:* Anticoagulation maintains extracorporeal circuit patency
- **Glucose:**
 - *Target:* >4.0 mmol/L, striving for 8-12 mmol/L
 - *Protocol:* Administer 2-4 mL of Glucose 50% (2.775 mmol/mL) per correction
 - *Rationale:* Active liver metabolism rapidly clears glucose. Ischemia-reperfusion injury (IRI) triggers cellular leakage, actively elevating perfusate glucose levels.
- **Sodium (Na⁺):**
 - *Target:* 135-145 mmol/L
 - *Protocol:* Administer NaCl 5% (0.86 mmol/mL) solely in the absence of NaHCO₃ administration. Target exactly 135 mmol/L. Initiate correction strictly after T10.
 - *Rationale:* Robust cellular metabolism utilizes active Na/K channels to pump sodium outward, accumulating sodium in the extracellular space.
- **Potassium (K⁺):**
 - *Target:* 3.5-5.0 mmol/L.
 - *Protocol:* Administer KCl (1 mmol/mL) specifically for verified hypokalemia. Apply strict limits to avoid over-correction.
 - *Rationale:* Robust cellular metabolism utilizes active Na/K channels to pump potassium inward, depleting potassium in the extracellular space.
- **Chloride (Cl⁻):**
 - *Target:* >90 mmol/L.
 - *Protocol:* Administer NaCl 5% (0.86 mmol/mL).
 - *Rationale:* Alkali bile production heavily relies on chloride. The CFTR channel actively drives transmembrane chloride into the bile. The AE2 membrane protein actively exchanges chloride for HCO₃ within the bile duct.
- **Calcium (Ca²⁺):**
 - *Target:* >0.70 mmol/L.
 - *Protocol:* Administer 10 mL Calcium gluconate (0.225 mmol/mL) at the initiation of COR. Administer a subsequent 10 mL during the normothermic phase. Proceed with granular 5 mL correction increments.
 - *Rationale:* Calcium functions as a critical cofactor driving continuous bile production.
- **Acid-Base Balance (pH & PCO₂):**
 - *Target:* pH <7.30 during COR, pH 7.35-7.45 during NMP. PCO₂ 4.5-6.0 kPa.
 - *Protocol:* Administer NaHCO₃ 8.4% (1 mmol/mL) to directly neutralize acid loads . Increase gas flow strictly upon confirmation of hepatic CO₂ production. Elevated airflow mechanically accelerates PCO₂ clearance, driving systemic pH upward .
 - *Rationale:* Anaerobic metabolism generates lactate, directly lowering pH at NMP onset. A mildly acidic environment provides active cytoprotection against IRI .

PCO₂ acts as an acidic metabolic byproduct and an active substrate for HCO₃ generation.

- **Oxygenation (PO₂):**

- *Target:* Maintain continuous levels >10 kPa.
- *Protocol:* Elevate FiO₂ concomitantly with increased gas flow.
- *Rationale:* Aerobic metabolism continuously consumes oxygen. Hypoxia drives reactive oxygen species production, inducing profound cellular toxicity.